Introduction to Modern Physics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Modern Physics (A24641)

Short Title:	Introduction to Modern Physics		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	CORAIFEARTAIGH		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces the student to fundamental concepts in quantum theory, relativity and cosmology. The first part of the module describes the emergence of the quantum in the description of phenomena such as blackbody radiation, the photoelectric effect, and atomic spectroscopy. The development of quantum theory is then described, from the hypothesis of wave-particle duality to the development of the Schrodinger equation. The role of quantum theory in solid-state physics is also described. The second part of the module concerns basic concepts in modern cosmology. The material covers the discovery of the expanding universe, the development of the big bang model and recent puzzles concerning the model. There will be a strong emphasis placed on problem-solving, logical thinking, and analysis throughout the module.

Programmes

`PHYI-0040`	BSc (Common Entry) (`WD_SSCIE_D`)	1 / 2 / E
`PHYI-0040`	BSc (Hons) in Food Science and Innovation (`WD_SFSIN_B`)	1 / 2 / E
`PHYI-0040`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	1 / 2 / E
`PHYI-0040`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	1 / 2 / E
`PHYI-0040`	BSc (Hons) in Science (Common Entry) (`WD_SSCCM_B`)	1 / 2 / E

Indicative Content

- Atomic physics: blackbody radiation, the photoelectric effect, atomic spectra, models of the atom
- New quantum theory: the Compton effect, the de Broglie hypothesis, the Schrodinger equation, the Heisenberg uncertainty principle
- Introduction to solid state physics: band theory, intrinsic and extrinsic semiconductors, p-n junction diode and rectification
- Relativity: space, time, matter and the general theory of relativity
- Cosmology: the expanding universe, the big bang model, today's lambda-CDM model of the universe

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Articulate the fundamental principles of the concept of the quantum.
2. Describe the fundamental principles of quantum theory.
3. Articulate the fundamental principles of solid state physics.
4. Express the fundamental principles of the big bang model of the universe.
5. Apply problem-solving skills to find solutions to simple problems in quantum physics, solid state physics and cosmology.

Learning and Teaching Methods

- Lectures: The lectures will be used to present new topics and their related concepts. Problem-solving techniques will also be used in class to analyse physical problems
- Tutorials: Tutorials will be used to give explanations of fundamental concepts in quantum physics and cosmology
- Self-directed study

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5
Essay	25%	4,5
Presentation	25%	4,5
In-Class Assessment	50%	1,2,3

Assessment Criteria

<40%: Inability to describe the fundamental principles of quantum theory, solid-state physics and cosmology.

40-49%: Ability to describe the fundamentals of quantum theory, solid-state physics and cosmology. Able to provide partial solutions to numerical problems.

50-59%: Ability to clearly describe and define concepts of the module. Able to analyse physical situations and apply suitable problem-solving techniques to find numerical solutions.

60-69%: IAbility to demonstrate very good knowledge of the module content, and present arguments relating to this content.

70%: All previous to an excellent level

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Independent Learning	87	

Essential Material

- Coles, P. _Cosmology: A very Short Introduction_. Oxford: Oxford University Press, 2001.
- Giancoli, D. _Principles with Applications_. 7th. New Jersey: Pearson, 2015.

Supplementary Material

- Halliday, D., R. Resnick and J. Walker. _Fundamentals of physics_. 10th. New York: Wiley, 2013.
- Stannard, R. _Relativity: A Very Short Introduction_. Oxford: Oxford University Press, 2008.

Requested Resources

- SCIENCE LAB: Physics